

Internship Project — Week 2

Project Title: Employee Attrition Prediction using Machine Learning

Write a short paragraph (6–10 lines) inside your notebook answering:

1. Which 3 factors most strongly predict that an employee will leave?

The three factors that most strongly predict an employee will leave are Monthly Income, the presence of overtime, and their Total Working Years (their overall career tenure and age).

2. Which department or job role should HR prioritize for retention efforts?

HR should prioritize the Sales department for immediate retention efforts, with a specific focus on the Sales Representative job role, which clearly shows the highest flight risk.

3. Does salary alone explain attrition, or are there other stronger factors?

Salary is a major driver, but it does not explain attrition on its own. Burnout indicators such as constant overtime, frequent business travel, and low Work-Life Balance ratings are extremely strong secondary factors that push even adequately paid employees to leave.

4. Write 2 concrete HR recommendations a company could implement based on your findings.

Recommendation 1: Implement mandatory retention check-ins at the critical 1-year and 5-year tenure marks, specifically targeting Sales Representatives.

Recommendation 2: Audit over time assignments across departments to identify overworked employees and enforce workload balancing to prevent burnout.

5. What limitation does this model have that an HR team should be aware of before using it?

A major limitation of this model is that it cannot account for unpredictable external human factors, such as a spouse relocating or a sudden health issue. Because of this, the model will inevitably have some false negatives, meaning it will miss some exiting employees because their internal company data looked completely normal.